

Calculus I Lesson 16C

1) $f(x) = x^2 + 3x + 5$

Find equation of the tangent line passing through the point $(-3, 5)$.

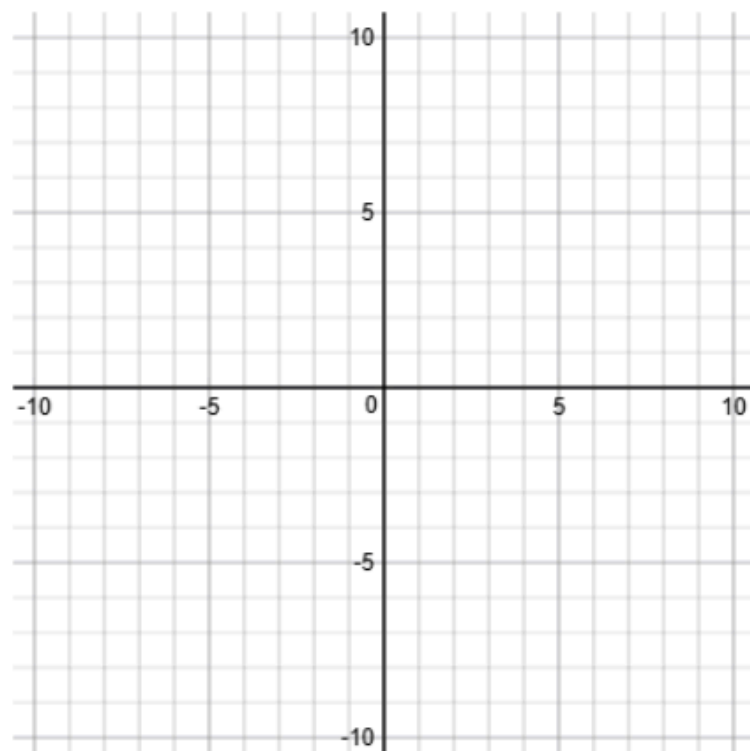
a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f'(-3) = \underline{\hspace{2cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = x^2 + 3x + 5$ and
the tangent line at the point $(-3, 5)$.



2) $f(x) = 4x^2 - 7x + 1$

Find equation of the tangent line passing through the point $(2, 3)$.

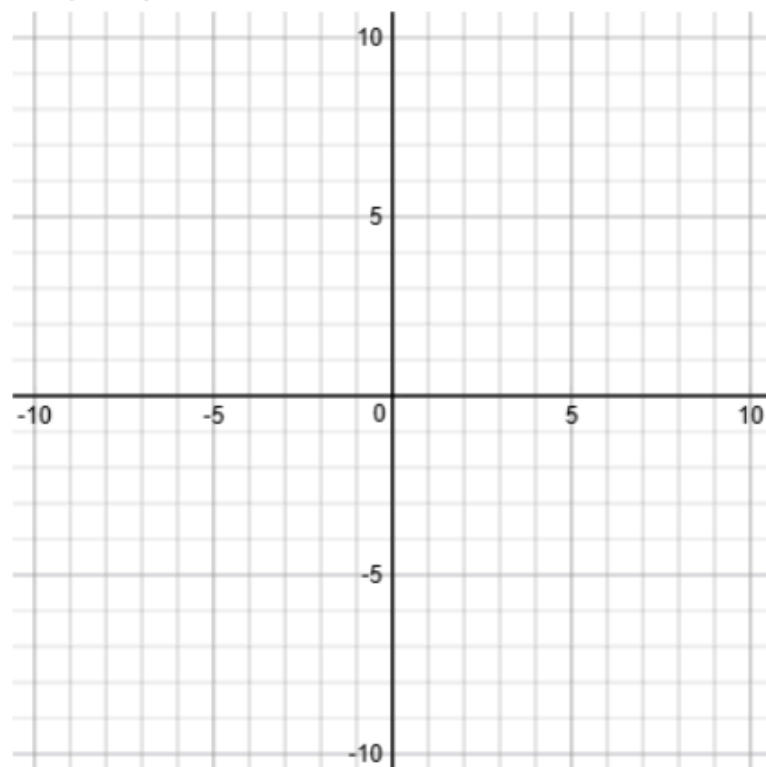
a) $f'(x) = \underline{\hspace{1cm}}?$

b) $f'(2) = \underline{\hspace{1cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = 4x^2 - 7x + 1$ and
the tangent line at the point $(2, 3)$.



3) $f(x) = x^4 + 3x^2 + 10x - 1$

Find equation of the tangent line passing through the point $(-1, -7)$.

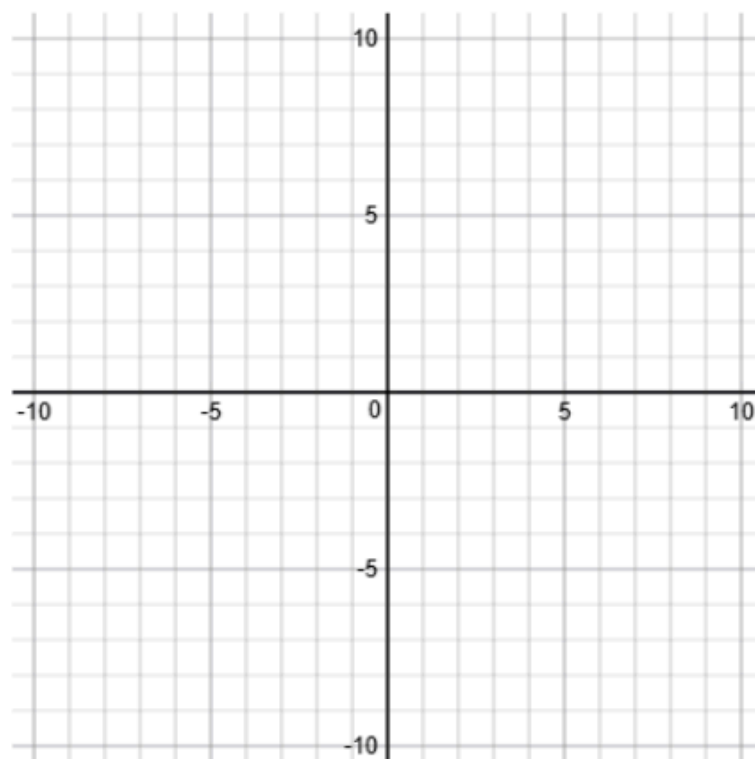
a) $f'(x) = \underline{\hspace{1cm}} ?$

b) $f'(-1) = \underline{\hspace{1cm}} ?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}} ?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = x^4 + 3x^2 + 10x - 1$ and
the tangent line at the point $(-1, -7)$.



4) $f(x) = 4x^{-2} + 13x^2 + 5x - 11$

Find equation of the tangent line passing through the point $(-1, 1)$.

a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f'(-1) = \underline{\hspace{2cm}}?$ This is the slope of the tangent line.

c) Equation of tangent line is: $\underline{\hspace{2cm}}?$

Hint: Use $y - y_1 = m(x - x_1)$

d) Graph $f(x) = 4x^{-2} + 13x^2 + 5x - 11$ and the tangent line at the point $(-1, 1)$.

